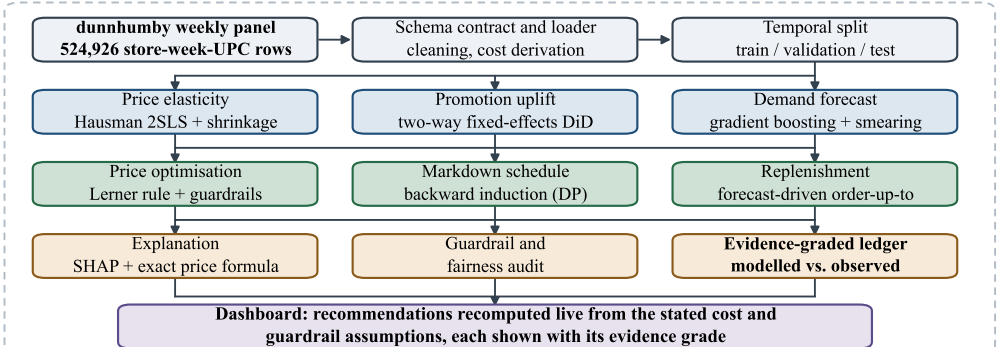
DATA
ESTIMATE
DECISION
ASSURE
DELIVER
dunnhumby weekly panel
524,926 store-week-UPC rows
Schema contract and loader
cleaning, cost derivation
Temporal split
train / validation / test
Price elasticity
Hausman 2SLS + shrinkage
Promotion uplift
two-way fixed-effects DiD
Demand forecast
gradient boosting + smearing
Price optimisation
Lerner rule + guardrails
Markdown schedule
backward induction (DP)
Replenishment
forecast-driven order-up-to
Explanation
SHAP + exact price formula
Guardrail and
fairness audit
Evidence-graded ledger
modelled vs. observed
Dashboard: recommendations recomputed live from the stated cost and
guardrail assumptions, each shown with its evidence grade
The identical pipeline is re-run on a schema-matched synthetic replica whose elasticity and promotion lift are known by construction